

Aplicaciones Lineales

Sea $F: \mathbb{R}^2 \rightarrow \mathbb{R}$ una aplicación lineal, se sabe que $F(1, 1) = 3$ y $F(2, 3) = 1$. Determinar $F(x, y)$.

$$F\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = x \cdot F\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) + y \cdot F\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right)$$

$$F\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right) = F(e_1) + F(e_2) = 3 ; F\left(\begin{pmatrix} 2 \\ 3 \end{pmatrix}\right) = 2F(e_1) + 3F(e_2) = 1$$

$$\begin{cases} F(e_1) + F(e_2) = 3 \\ 2F(e_1) + 3F(e_2) = 1 \end{cases} \sim \left(\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & 3 & 1 \end{array}\right) \sim \left(\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & -5 \end{array}\right)$$

$$F(e_2) = -5 \quad F(e_1) = 3 + 5 = 8$$

$$F\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = x \cdot F(e_1) + y \cdot F(e_2) = x \cdot 8 + y \cdot (-5)$$

$$F\left[\begin{pmatrix} x \\ y \end{pmatrix}\right] = 8x - 5y$$

Sea $F: \mathbb{R}^{2 \times 2} \rightarrow \mathbb{P}_2$ una aplicación lineal, se sabe que $F\left[\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\right] = x$, $F\left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right] = x^2$, $F\left[\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}\right] = 1$ y $F\left[\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\right] = 0$. Determinar $F\left[\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right]$.

$$e_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad e_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad e_3 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \quad e_4 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$F\left(\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\right) = x \rightarrow F(e_1) + F(e_2) + F(e_3) + F(e_4) = x$$

$$F\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right) = x^2 \rightarrow F(e_1) + F(e_4) = x^2 ; F\left(\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}\right) = 1 \rightarrow 2F(e_1) + F(e_2) + F(e_4) = 1$$

$$F\left(\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\right) = 0 \rightarrow F(e_1) = 0 \quad F\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = a \cdot F(e_1) + b \cdot F(e_2) + c \cdot F(e_3) + d \cdot F(e_4)$$

$$\begin{cases} F(e_1) + F(e_2) + F(e_3) + F(e_4) = x \\ F(e_1) + F(e_4) = x^2 \\ 2F(e_1) + F(e_2) + F(e_4) = 1 \\ F(e_1) = 0 \end{cases} \quad \begin{cases} F(e_3) = x - x^2 + x^2 - 1 = x - 1 \\ F(e_4) = x^2 \\ F(e_2) = 1 - x^2 \\ F(e_1) = 0 \end{cases}$$

$$F\left[\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right] = a \cdot 0 + b \cdot (1 - x^2) + c \cdot (x - 1) + d \cdot x^2$$

3) Sea $F: \mathbb{P}_2 \rightarrow \mathbb{R}^2$ una aplicación lineal, se sabe que:
 $F(t^2 - 2t) = (1; 1)$
 $F(t) = (0; 2)$
 $F(t^2 + 1) = (-1; 3)$
 Determinar la expresión analítica de la aplicación F .

$$\mathbb{P}_2 = \text{lin} \left\{ \overset{e_1}{t^2}, \overset{e_2}{t}, \overset{e_3}{1} \right\}$$

$$F(t^2 - 2t) = (1; 1) \rightarrow F(e_1) - 2F(e_2) = (1; 1) \rightarrow F(e_1) = (1; 1) + (0; 2) \cdot 2$$

$$F(t) = (0; 2) \rightarrow F(e_2) = (0; 2)$$

$$F(e_1) = (1; 5)$$

$$F(t^2 + 1) = (-1; 3) \rightarrow F(e_1) + F(e_3) = (-1; 3) \rightarrow F(e_3) = (-1; 3) - (1; 5)$$

$$F(e_3) = (-2; -2)$$

$$F: \mathbb{P}_2 \rightarrow \mathbb{R}^2 \quad \mathbb{P}_2 \ni at^2 + bt + c = a \cdot F(e_1) + b \cdot F(e_2) + c \cdot F(e_3)$$

$$F(at^2 + bt + c) \rightarrow a \cdot \begin{pmatrix} 1 \\ 5 \end{pmatrix} + b \cdot \begin{pmatrix} 0 \\ 2 \end{pmatrix} + c \cdot \begin{pmatrix} -2 \\ -2 \end{pmatrix} = \begin{pmatrix} a - 2c \\ 5a - 2c \end{pmatrix}$$